

- 8 (a) Solar power collected = intensity x surface area

$$= (0.90 \times 10^3) \times 7.7 = 6930 \text{ W} \quad \text{M1}$$

Time taken to travel 200 km journey = distance / speed = 200 km / 67 km h⁻¹

$$= (200/67) \text{ h} = (200/67) \times 3600 \text{ s}$$

$$= 1.07 \times 10^4 \text{ s} \quad \text{M1}$$

$$\text{Total energy collected} = \text{power} \times \text{time} = 6930 \times (200/67) \times 3600 = 7.45 \times 10^7 \text{ J} \quad \text{A1}$$

Comments:

Power of ten error in some answers. E.g. Solar power collected = $(0.90) \times 7.7 = 6.93 \text{ W}$

- (b) electrical energy transformed = 26% of solar input energy

$$= (26/100) \times 7.45 \times 10^7$$

$$= 1.94 \times 10^7 \text{ J} \quad \text{M1}$$

$$= 5.39 \text{ kWh} \quad \text{A1}$$

Comments:

Some students thought kWh is a unit for power.

Many were unable to convert J to kWh correctly. Equating kWh to kJ is one of the common errors.

- (c) At average power,

$$\text{Efficiency} = (\text{motor power output} / \text{electrical power input}) \times 100\% = 92\% \quad \text{C1}$$

$$\text{Electrical input power by battery} = 1.1 / 0.92 \quad \text{M1}$$

$$= 1.2 \text{ kW}$$

A0

(d) (i) $N \times \text{area of each cell} = \text{total area of the cells}$

$$\Rightarrow N = 7.7 / 6.4 \times 10^{-3} = 1.2 \times 10^3$$

B1

(ii) $N \times \text{energy collected by each cell} = \text{total solar energy collected by all cells}$

$$\text{Energy collected by each cell} = 7.45 \times 10^7 / 1.2 \times 10^3$$

C1

$$= 6.2 \times 10^4 \text{ J}$$

B1

(e) Given: $R \propto v^2 \Rightarrow R = kv^2$ where $k = \text{constant}$

At constant speed, net force = 0

$\Rightarrow$ Driving force of motor $F = \text{total resistive force } R$

C1

Using $P = Fv = Rv = kv^2 v$

$$P = kv^3 \text{ i.e. } \langle P \rangle = k \langle v \rangle^3 \text{ and } P_{\text{peak}} = k v_{\text{peak}}^3$$

M1

$$\Rightarrow 1100 / 4500 = 67^3 / v_{\text{peak}}^3$$

$$\Rightarrow v_{\text{peak}} = 107 \text{ km h}^{-1}$$

A1

Comments:

Some students realised that at constant velocity, forward force and total resistive force are equal in magnitude. However, many did not go on to work out the correct relationship between power and velocity. A significant number of students wrongly equate $P = v^3$.

(f) Using $P = IV$,

$$I = P/V = 1200 / 50 = 24 \text{ A}$$

B1

(g) Using $Q = It$,

$$t = 60 \text{ A h} / 24 \text{ A}$$

M1

$$= 2.5 \text{ h}$$

A1

Comments:

Some students have difficulty with interpretation of charge capacity in Ah and calculated $1/t$ instead, using $24/60$.

(h) Any two factors:

1. weather conditions of the day – amount of cloud cover

2. Dirt on panel or shade by trees.

3. Angle of incidence of the solar beam on the solar cells

B2

Comments:

Answer should focus on specific factors affecting intensity of solar energy received by the solar cells.

- (i) 1. Wide to maximize surface area exposed to sunlight.
2. Flat to minimize energy losses due to air resistance.

B2

Comments:

Most students know maximizing sunlight exposure is key but attributed it to flatness rather than wide area.